

directions perpendicular to the line but slowly along its length. Coupler fields are associated with the coupler itself rather than the power line. Radiation fields are considered to be a source of interference because these fields decay relatively slowly.

Interference associated with BPL can be broadly categorised into two groups: near-field ($0.62(D^2/\lambda)^{1/2} < r < 2D^2/\lambda$) radiation and far-field ($r > 2D^2/\lambda$) radiation, where 'r' is the distance from the radiator, 'D' is the largest linear dimension of the radiator and ' λ ' is signal wavelength. Near-field effects persist along the entire length of the wire and may cause serious interference issues. Far-field effects can be quite troublesome because ionospheric propagation of HF radiation can travel thousands of kilometres. In order to reduce these radiations, a balanced configuration of two MV wires driven differentially and spaced appropriately can be employed.

Power line noise. MV/LV power lines are inherently very noisy due to changing nature of the load connected to power lines, number and types of line branches, length of line branches, type of power line equipment connected (such as capacitor banks and transformers), and impedance mismatch caused by unterminated stubs and line branches. In addition, on/off switching of capacitor banks used to correct power factor and switching power supplies often introduce noisy harmonics into the line.

Unlike twisted-pair (used in DSL) or shielded coaxial cable, power lines have no inherent noise cancellation mechanism. In order to cope up with noisy environment of power lines, OFDM along with a number of channel interference noise ratio based modulations such as QAM, BPSK and QPSK may be a solution. OFDM is very robust against frequency selectivity but any time-varying characteristic of the channel limits system performance. Time variations deteriorate the orthogonality of the sub-carriers, resulting in inter-carrier interference. To eliminate inter-carrier interference, a guard time is inserted with a length

longer than the duration of the impulse response of the channel. The insertion of guard time has the penalty of a loss in the signal-to-noise ratio that further increases the bandwidth requirement.

Bandwidth issues. Each distribution transformer feeds power to 10 to 50 houses/offices. Since bandwidth to the transformer is limited and the same power line is shared by all these houses, there is scarcity of bandwidth available to each customer. Moreover, BPL is a contention-based system (CSMA/CA), which again imposes additional challenges. Similar to DSL, speeds (bit-rates) of BPL systems available to customers are also dependent on the distance between the supplying substation and the customer's home.

Security issues. BPL signal propagates in a LAN-like manner, which makes detection and interception of neighbouring transmissions simple. So in order to prevent interception of legitimate customers' data by unauthorised intruders, strong authentication, authentication and data encryption algorithms such as DES and AES should be used.

BPL in a nutshell

BPL technology is a union of two applications in a single system, holding great promise as a ubiquitous broadband solution that would offer a viable alternative to cable, digital subscriber line, fibre and wireless broadband solutions. Additionally, it offers the ability to employ intelligent power line networks that make use of supervisory control and data acquisition devices. As standardisation, interference mitigation and improvement in technology are on their way, the future of BPL looks very bright. ●

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